

Constructions of Efficiently Implementable Boolean Functions with Provable Nonlinearity/Resiliency/Algebraic Immunity Trade-Offs

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Abstract

We describe several families of efficiently implementable Boolean functions achieving provable trade-offs between resiliency, nonlinearity, and algebraic immunity. In particular, the following statement holds for each of the function families that we propose. Given integers $m_0 \geq 0$, $x_0 \geq 1$, and $a_0 \geq 1$, it is possible to construct an n -variable function which has resiliency at least m_0 , linear bias (which is an equivalent method of expressing nonlinearity) at most 2^{-x_0} and algebraic immunity at least a_0 ; further, n is linear in $\max(m_0, x_0, a_0)$, and the function can be implemented using $O(n)$ 2-input gates, which is essentially optimal.

Keywords: Boolean function, resiliency, nonlinearity, algebraic immunity, efficient implementation.

1 Introduction

Boolean functions have widespread applications in various areas of computer science and engineering. For cryptographic applications, certain properties of Boolean functions have been identified as necessary for providing resistance to known attacks. Three such extensively studied properties are resiliency, nonlinearity and algebraic immunity. Over the last few decades extensive research has been carried out on various aspects of Boolean functions possessing one or more of these three properties. We refer to the excellent book [4] for a very comprehensive and unified treatment of cryptographic properties of Boolean functions.

It is easy to construct functions which maximise any one of the three properties of resiliency, nonlinearity and algebraic immunity. For an n -variable function, the maximum possible order of resiliency is $n - 1$ and the only functions which achieve this order of resiliency are the two affine functions which are non-degenerate on all the n variables. For even n , the maximum possible nonlinearity is achieved by bent functions [18] and there are many well known constructions of bent functions. The maximum possible algebraic immunity [9] of an n -variable function is $\lceil n/2 \rceil$, and the majority function achieves this value of algebraic immunity [10]. Functions maximising one of the properties usually have poor behaviour with respect to the other two properties. Affine functions have minimum nonlinearity and algebraic immunity, bent functions are not balanced (i.e. not 0-resilient), while the majority function has poor nonlinearity and resiliency.

This brings up the issue of trade-offs between these properties. There are two aspects to such trade-offs. The first aspect is that of determining the exact nature of the trade-off curve, and the second

aspect is that of obtaining construction methods for functions which achieve a desired trade-off. Both of these are very difficult questions and progress on answering these questions have been very slow.

While resiliency, nonlinearity and algebraic immunity are security properties of Boolean functions, there is another aspect of Boolean functions which is also of great practical importance. For use in actual design of cryptographic systems, it is crucial that any component Boolean functions is efficient to implement. A measure of implementation efficiency is the number of 2-input gates that is required to implement a Boolean function. From a cryptographic point of view, along with good security properties, a Boolean function also needs to have a low gate count. Keeping the efficiency aspect in mind, the challenge is the following.

Challenge: Obtain constructions of infinite families of Boolean functions with provable values or bounds on resiliency, nonlinearity, and algebraic immunity, such that the functions can be efficiently implemented.

In this paper, we provide the first general answers to the above challenge. See below for a summary of known partial results. Instead of nonlinearity, we describe our results in terms of the equivalent notion of linear bias. We describe several infinite families of Boolean functions for which the following strong result holds.

Theorem (informal): Given integers $m_0 \geq 0$, $x_0 \geq 1$ and $a_0 \geq 1$, it is possible to construct an n -variable function which has resiliency at least m_0 , linear bias at most 2^{-x_0} and algebraic immunity at least a_0 , where n is linear in $\max(m_0, x_0, a_0)$, and the function can be implemented using $O(n)$ 2-input gates, which is asymptotically optimal in the sense that any function which is at least m_0 -resilient, has linear bias at most 2^{-x_0} , and algebraic immunity at least a_0 must require $\Omega(\max(m_0, x_0, a_0))$ 2-input gates.

Our construction leverages a recent result from [7] which showed how to construct a special class of bent functions with provable lower bound on algebraic immunity. Two of the infinite classes that we describe are obtained by extending such bent functions in a simple manner. The first class simply adds a number of new variables, while the second class adds a 5-variable, 1-resilient function and then adds a number of new variables. We show that the above mentioned strong result holds for both of these classes.

The addition of new variables increases resiliency. Simply adding a significant number of new variables to increase resiliency may not be completely desirable. We describe two more classes of functions which are not obtained by just adding new variables to bent functions. To obtain these new classes we resurrect a two-and-half decades old sketch of an idea from [17]. By suitably fleshing out the idea with complete details and proofs, we construct families of functions for which the above strong result holds and further for which the functions are not obtained by adding new variables.

Previous and related works

To the best of our knowledge, the above mentioned challenge has not been addressed in its full generality in the literature. Partial results are known. Below we mention these partial results and compare to our results.

Provable algebraic immunity/linear bias trade-offs for balanced (i.e. 0-resilient) functions.

Various constructions have been proposed [5, 27, 3, 22, 20, 28, 13] in the literature which provide functions with maximum algebraic immunity and provable upper bound on linear bias for balanced (i.e. 0-resilient) functions. In contrast, we provide functions which achieve almost optimal linear bias and algebraic immunity which is at least half the maximum possible value. The upper bound on the linear bias, as well as the actual values of linear bias for concrete functions, obtained from the previously proposed constructions are higher than the linear bias of the functions that we construct.

Provable algebraic immunity/linear bias trade-offs for 1-resilient functions. For even n , previous works [26, 29, 23, 30, 24] have proposed constructions of 1-resilient functions with maximum algebraic immunity and provable upper bound on the linear bias. Till date, the last of this line of work is [24] which provides functions with lower linear bias compared to all previous works. The work [24] also provides a lower bound on the fast algebraic immunity of the constructed functions. In comparison, for even n , the 1-resilient functions that we construct achieve almost optimal linear bias which is lower than the linear bias (both upper bound and actual values for concrete functions) of the functions constructed in [24], but the algebraic immunity is about half the maximum possible value, which also guarantees that the fast algebraic immunity is about a quarter of the maximum possible value.

So for both the cases of balancedness and 1-resiliency, the previously proposed functions and the functions that we propose achieve different points on the algebraic immunity/linear bias trade-off curve. From the point of efficiency, however, the functions that we construct require $O(n)$ gates for n -variable functions, while the previous functions essentially require discrete logarithm computation and hence require super-polynomial size circuits. In concrete terms, this has a very important effect. Our constructions can be easily scaled up to achieve target values of algebraic immunity, fast algebraic immunity and linear bias. Later we provide concrete examples to illustrate the importance of scalability.

Provable algebraic immunity/linear bias trade-offs for m -resilient functions with $m > 1$.

As far as we are aware, there is no previous work in the literature which provides provable algebraic immunity/linear bias trade-off for m -resilient functions with $m > 1$. So for $m > 1$, we provide the first constructions of such functions.

Provable algebraic immunity/resiliency trade-off. From the viewpoint of theoretical computer science, the properties of algebraic immunity and resiliency were shown [1] to be key parameters of the “local function” required in Goldreich’s construction [12] of pseudorandom generators. Motivated by [1], a recent study [11] dealt quite extensively with the provable trade-off between resiliency and algebraic immunity. This line of work does not consider nonlinearity (or equivalently linear bias) of the functions. So while the algebraic immunity/resiliency trade-off is of theoretical interest, it is perhaps of limited relevance to the context of cryptographic applications of Boolean functions.

Outline of the paper

The background and preliminaries are described in Section 2. In Section 3, we present the two basic constructions. The iterated construction is presented in Section 4. Based on the constructions in the previous sections, the main result is presented in Section 5. The concluding remarks are provided in Section 6.

2 Background and Preliminary Results

In this section, we provide the basic definitions and present some preliminary results. For further and extensive details on cryptographic properties of Boolean functions we refer to [4].

The cardinality of a finite set S will be denoted by $\#S$. $\mathbb{F}_2$ denotes the finite field of two elements; $\mathbb{F}_2^n$, where n is a positive integer, denotes the vector space of dimension n over $\mathbb{F}_2$. The addition operator over both $\mathbb{F}_2$ and $\mathbb{F}_2^n$ will be denoted by $\oplus$. The product (which is also the logical AND) of $x, y \in \mathbb{F}_2$ will simply be written as xy . A bit vector of dimension n , i.e. an element of $\mathbb{F}_2^n$ will also be considered to be an n -bit binary string.

Let n be a positive integer. The support of $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_2^n$ is $\text{supp}(\mathbf{x}) = \{1 \leq i \leq n : x_i = 1\}$, and the weight of $\mathbf{x}$ is $\text{wt}(\mathbf{x}) = \#\text{supp}(\mathbf{x})$. By $\mathbf{0}_n$ and $\mathbf{1}_n$ we will denote the all-zero and all-one strings of length n respectively. For $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$, with $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ the distance between $\mathbf{x}$ and $\mathbf{y}$ is $d(\mathbf{x}, \mathbf{y}) = \#\{i : x_i \neq y_i\}$; the inner product of $\mathbf{x}$ and $\mathbf{y}$ is $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 \oplus \dots \oplus x_ny_n$.

Boolean function. An n -variable Boolean function f is a map $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$. The weight of f is $\text{wt}(f) = \#\{\mathbf{x} \in \mathbb{F}_2^n : f(\mathbf{x}) = 1\}$; f is said to be *balanced* if $\text{wt}(f) = 2^{n-1}$. The canonical ordering of the elements of $\mathbb{F}_2^n$ is the ordering where for $0 \leq i < 2^n$, the i -th element in the ordering is the n -bit binary representation of i . With respect to the canonical ordering an n -variable function f can be represented by a binary string of length 2^n , where the i -th bit of the string is the value of f on the i -th element of the canonical representation. We call such a string to be the *string (or truth table) representation* of f .

The *algebraic normal form (ANF) representation* of an n -variable Boolean function f is the representation of f as an element of the polynomial ring $\mathbb{F}_2[X_1, \dots, X_n]/(X_1^2 \oplus X_1, \dots, X_n^2 \oplus X_n)$ in the following manner: $f(X_1, \dots, X_n) = \bigoplus_{\alpha \in \mathbb{F}_2^n} a_\alpha \mathbf{X}^\alpha$, where $\mathbf{X} = (X_1, \dots, X_n)$; for $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{F}_2^n$, $\mathbf{X}^\alpha$ denotes the monomial $X_1^{\alpha_1} \dots X_n^{\alpha_n}$; and $a_\alpha \in \mathbb{F}_2$. The (algebraic) degree of f is $\deg(f) = \max\{\text{wt}(\alpha) : a_\alpha = 1\}$; we adopt the convention that the zero function has degree 0. The degree (or sometimes also called the length) of the monomial $\mathbf{X}^\alpha$ is $\text{wt}(\alpha)$.

It is useful to introduce a notation for the concatenation of two functions.

Construction 1 Let g and h be n -variable functions. Define an $(n+1)$ -variable function in the following manner.

$$f(X_1, \dots, X_{n+1}) = (1 \oplus X_{n+1})g(X_1, \dots, X_n) \oplus X_{n+1}h(X_1, \dots, X_n).$$

We denote f as $\text{Concat}(g, h)$.

Note that the string representation of $f = \text{Concat}(g, h)$ is obtained by concatenating the string representations of g and h .

An n -variable function $f(X_1, \dots, X_n)$ is said to be *non-degenerate* on the variable X_i , $1 \leq i \leq n$, if there are $\alpha, \beta \in \mathbb{F}_2^n$ which differ only in the i -th position and $f(\alpha) \neq f(\beta)$; if there are no such α and β , then f is said to be *degenerate* on the variable X_i . The following result provides a lower bound on the number of 2-input gates required to compute a non-degenerate function.

Proposition 1 (Proposition 1 of [31]) Any circuit for an n -variable non-degenerate function consists of at least $n - 1$ 2-input gates.

Functions of degree at most 1 are said to be affine functions. Affine functions with $a_{\mathbf{0}_n} = 0$ are said to be linear functions. Each $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{F}_2^n$, defines the linear function $\langle \alpha, \mathbf{X} \rangle =$

$\langle \alpha, (X_1, \dots, X_n) \rangle = \alpha_1 X_1 \oplus \dots \oplus \alpha_n X_n$. If $\text{wt}(\alpha) = w$, then the function $\langle \alpha, \mathbf{X} \rangle$ is non-degenerate on exactly w of the n variables.

Nonlinearity and Walsh transform. The distance between two n -variable functions f and g is $d(f, g) = \#\{\mathbf{x} \in \mathbb{F}_2^n : f(\mathbf{x}) \neq g(\mathbf{x})\} = \text{wt}(f \oplus g)$. The *nonlinearity* of an n -variable function f is defined to be $\text{nl}(f) = \min_{\alpha \in \mathbb{F}_2^n} \{d(f, \langle \alpha, \mathbf{X} \rangle), d(f, 1 \oplus \langle \alpha, \mathbf{X} \rangle)\}$, i.e. the nonlinearity of f is the minimum of the distances of f to all the affine functions.

The Walsh transform of an n -variable function f is the map $W_f : \mathbb{F}_2^n \rightarrow \mathbb{Z}$, where for $\alpha \in \mathbb{F}_2^n$, $W_f(\alpha) = \sum_{\mathbf{x} \in \mathbb{F}_2^n} (-1)^{f(\mathbf{x}) \oplus \langle \alpha, \mathbf{x} \rangle}$. From the definition it follows that $W_f(\alpha) = 2^n - 2d(f, \langle \alpha, \mathbf{X} \rangle)$. Consequently, $\text{nl}(f) = 2^{n-1} - \frac{1}{2} \max_{\alpha \in \mathbb{F}_2^n} |W_f(\alpha)|$. The nonlinearity of f is invariant under an invertible linear transformation on the variables of f .

We define the *linear bias* of an n -variable function f to be $\text{LB}(f) = \max_{\alpha \in \mathbb{F}_2^n} |W_f(\alpha)|/2^n = 1 - \text{nl}(f)/2^{n-1}$. From a cryptographic point of view, the linear bias, rather than the nonlinearity, is of importance, since it is the linear bias which is used to quantify the resistance to (fast) correlation attacks.

Bent functions. An n -variable function f is said to be bent [18] if $W_f(\alpha) = \pm 2^{n/2}$ for all $\alpha \in \mathbb{F}_2^n$. From the definition it follows that bent functions can exist only if n is even. An n -variable bent function has nonlinearity $2^{n-1} - 2^{n/2-1}$ (resp. linear bias $2^{-n/2}$), and this is the maximum possible nonlinearity (resp. least possible linear bias) that can be achieved by any n -variable function.

The well known Maiorana-McFarland class of bent functions is defined as follows. For $k \geq 1$, let $\psi : \{0, 1\}^k \rightarrow \{0, 1\}^k$ be a bijection and $h : \{0, 1\}^k \rightarrow \{0, 1\}$ be a Boolean function. Let $\mathbf{X} = (X_1, \dots, X_k)$ and $\mathbf{Y} = (Y_1, \dots, Y_k)$. For $k \geq 1$, (ψ, h) -MM $_{2k}$ is defined to be the following function.

$$(\psi, h)\text{-MM}_{2k}(\mathbf{X}, \mathbf{Y}) = \langle \psi(\mathbf{X}), \mathbf{Y} \rangle \oplus h(\mathbf{X}). \quad (1)$$

Note that the degree of (ψ, h) -MM $_{2k}$ is $\max(1 + \max_{1 \leq i \leq k} \deg(\psi_i), \deg(h))$, where $\psi_1, \dots, \psi_k$ are the component functions of ψ .

Almost optimal linear bias. For a positive integer n , the covering radius bound (see Theorem 3 in Chapter 3 of [4]) on the nonlinearity of an n -variable function f is the following: $\text{nl}(f) \leq 2^{n-1} - \lfloor 2^{n/2-1} \rfloor$, and equivalently, $\text{LB}(f) \geq \lfloor 2^{n/2-1} \rfloor / 2^{n-1}$. The bound is achieved if and only if f is bent. Let $\chi(n) = \lfloor 2^{n/2-1} \rfloor / 2^{n-1}$. We say that f has *almost optimal linear bias* if $\chi(n) \leq \text{LB}(f) \leq 2\chi(n)$, i.e. if the linear bias of f is at most two times the lower bound arising from the covering radius bound. If n is even, and $\text{nl}(f) = 2^{n-1} - 2^{n/2}$, then $\text{LB}(f) = 2^{-(n-2)/2} = 2\chi(n)$, while if n is odd, and $\text{nl}(f) = 2^{n-1} - 2^{(n-1)/2}$, then $\text{LB}(f) = 2^{-(n-1)/2} < 2\chi(n)$; in both cases the linear bias is almost optimal.

Resilient functions. Let n be a positive integer and m be an integer such that $0 \leq m < n$. An n -variable function f is said to be m -resilient if $W_f(\alpha) = 0$ for all α satisfying $\text{wt}(\alpha) \leq m$. Equivalently, f is m -resilient if and only if $d(f, \langle \alpha, \mathbf{X} \rangle) = 2^{n-1}$ for α satisfying $\text{wt}(\alpha) \leq m$, i.e. if the distance between f and any linear function which is non-degenerate on at most m variables is equal to 2^{n-1} . Siegenthaler's bound [21] relates n , m and the degree d of f in the following manner:

$$\text{if } m = n - 1, \text{ then } d = 1, \text{ and if } m \leq n - 2, \text{ then } d \leq n - m - 1. \quad (2)$$

Divisibility results obtained in [19, 2, 6] show that the Walsh transform values of an n -variable, m -resilient function f having degree d is divisible by $2^{m+2+\lfloor(n-m-2)/d\rfloor}$.

Suppose $f(W, \mathbf{X})$ is defined to be $f(W, \mathbf{X}) = W \oplus g(\mathbf{X})$. Then $\text{nl}(f) = 2 \cdot \text{nl}(g)$ and $\text{LB}(f) = \text{LB}(g)$. Further, f is balanced, and if g is m -resilient, then f is $(m+1)$ -resilient.

Algebraic immunity. The algebraic immunity of an n -variable function f is defined [9, 16] as follows: $\text{AI}(f) = \min_{g \neq 0} \{\deg(g) : \text{either } gf = 0, \text{ or } g(f \oplus 1) = 0\}$. It is known [9] that $\text{AI}(f) \leq \lceil n/2 \rceil$. If the bound is achieved, then we say that f has optimal algebraic immunity.

Fast algebraic immunity. Given an n -variable function f , suppose that there are n -variable functions $g \neq 0$ and h of degrees e and d respectively such that $gf = h$. If $e + d \geq n$, then the existence of g and h satisfying $gf = h$ is guaranteed [8]. The fast algebraic immunity (FAI) of f is defined in the following manner: $\text{FAI}(f) = \min(2\text{AI}(f), \min_{g \neq 0} \{\deg(g) + \deg(fg) : 1 \leq \deg(g) < \text{AI}(f)\})$. The following bounds hold for $\text{FAI}(f)$: $1 + \text{AI}(f) \leq \text{FAI}(f) \leq 2\text{AI}(f)$. In particular, the lower bound of $1 + \text{AI}(f)$ on $\text{FAI}(f)$ suggests that a target value (which is not necessarily optimal) of $\text{FAI}(f)$ may be achieved by designing functions with a desired value of algebraic immunity.

Majority function. For $n \geq 1$, let $\text{Maj}_n : \{0, 1\}^n \rightarrow \{0, 1\}$ be the majority function defined in the following manner. For $\mathbf{x} \in \{0, 1\}^n$, $\text{Maj}(\mathbf{x}) = 1$ if and only if $\text{wt}(\mathbf{x}) > \lfloor n/2 \rfloor$.

Theorem 1 (Theorems 1 and 2 of [10]) *Let n be a positive integer.*

1. Maj_n has the maximum possible AI of $\lceil n/2 \rceil$.
2. The degree of Maj_n is equal to $2^{\lfloor \log_2 n \rfloor}$.

Proposition 2 (Proposition 7 of [7]) Maj_n can be implemented using $O(n)$ NAND gates.

Direct sum. A simple way to construct a function is to add together two functions on disjoint sets of variables. The constructed function is called the direct sum of the two smaller functions. Let n_1 and n_2 be positive integers and g and h be functions on n_1 and n_2 variables respectively. Define

$$f(X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2}) = g(X_1, \dots, X_{n_1}) \oplus h(Y_1, \dots, Y_{n_2}). \quad (3)$$

Bounds on the algebraic immunity of a function constructed as a direct sum is given by the following result.

Proposition 3 (Lemma 3 of [15]) *For f constructed as in (3), $\max\{\text{AI}(g), \text{AI}(h)\} \leq \text{AI}(f) \leq \text{AI}(g) + \text{AI}(h)$.*

Maiorana-McFarland with Majority. A lower bound on the algebraic immunity of a special class of Maiorana-McFarland bent functions was obtained in [7].

Theorem 2 (Theorem 2 of [7]) *Let $k \geq 2$, $n = 2k$, and $\psi : \{0, 1\}^k \rightarrow \{0, 1\}^k$ be an affine map, i.e. each of the coordinate functions of ψ is an affine function of the input variables. Then*

$$\text{AI}((\psi, h)\text{-MM}_{2k}) \geq \text{AI}(h). \quad (4)$$

Consequently, $\text{AI}((\psi, \text{Maj})\text{-MM}_{2k}) \geq \text{AI}(\text{Maj}_k) = \lceil k/2 \rceil$.

Theorem 2 holds for any affine map ψ . From efficiency considerations, we will consider ψ to be a bit permutation, i.e. there is a permutation ρ of $\{1, \dots, k\}$ such that for any $(x_1, \dots, x_k) \in \mathbb{F}_2^n$, $\psi(x_1, \dots, x_k) = (x_{\rho(1)}, \dots, x_{\rho(k)})$. Implementation of a bit permutation requires only the proper connection pattern, and does not require any gates. For the case where ψ is a bit permutation, it was shown in [14] that the algebraic immunity of $(\psi, \text{Maj})\text{-MM}_{2k}$ is at most $1 + 2^{\lceil \log(n/4) \rceil}$.

Gate count. From an implementation point of view, it is of interest to obtain functions which are efficient to implement. A measure of implementation efficiency is the number of 2-input gates s required to implement an n -variable function f . The truth table representation of f requires $s = \Omega(2^n)$. For even moderate values of n , such a representation results in a very large circuit. From an implementation point of view, it is of interest to obtain functions f where $s = O(n)$, which in view of Proposition 1 is asymptotically optimal. In this paper, we consider the following 2-input gates: XOR, AND, OR, and NAND.

3 Basic Constructions of m -Resilient Functions

In the present section, we provide two basic constructions of m -resilient functions with guarantees on linear bias and algebraic immunities.

The following result builds on the basic fact that adding new variables increases the order of resiliency.

Theorem 3 *Let m be a non-negative integer, and $n > m$ be another integer such that $n \not\equiv m \pmod{2}$. Let $k = (n - m - 1)/2$. Let $\psi : \{0, 1\}^k \rightarrow \{0, 1\}^k$ be a bit permutation. Define*

$$\begin{aligned} f(X_1, \dots, X_{m+1}, U_1, \dots, U_k, V_1, \dots, V_k) \\ = X_1 \oplus \dots \oplus X_{m+1} \oplus (\psi, h)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k). \end{aligned} \quad (5)$$

Then f is an n -variable, m -resilient function with linear bias equal to $2^{-(n-m-1)/2}$. Further, if $h = \text{Maj}_k$, then the algebraic immunity of f is at least $\lceil (n - m - 1)/4 \rceil$, and f can be implemented using $O(n)$ gates.

Proof: Since $m+1$ new variables are added to a bent function, the order of resiliency is m . The linear bias of the bent function on $2k$ variables is 2^{-k} , where $k = (n - m - 1)/2$, and since the new variables are simply added, the linear bias remains unchanged. From Proposition 3 the algebraic immunity of f is at least the algebraic immunity of the bent function; from Theorem 2 the algebraic immunity of the bent function is at least the algebraic immunity of $h = \text{Maj}_k$; and from Theorem 1 the algebraic immunity of Maj_k is equal to $\lceil k/2 \rceil$. From Proposition 2, Maj_k can be implemented using $O(k) = O(n)$ gates. The implementation of the other components of f also require $O(n)$ gates. $\square$

The special case of balanced functions was considered in [7] and is given in the following result.

Corollary 1 ([7]) *Let $n \equiv 1 \pmod{2}$. It is possible to construct an n -variable, balanced function with linear bias equal to $2^{-(n-1)/2}$ (which is almost optimal), algebraic immunity at least $\lceil (n - 1)/4 \rceil$, and can be implemented using $O(n)$ gates.*

Proof: Putting $m = 0$ in Theorem 3 provides the result. $\square$

Several papers [5, 27, 3, 22, 20, 28, 13] proposed constructions of n -variable balanced functions which achieve optimal algebraic immunity $\lceil n/2 \rceil$ with provable upper bounds on the linear bias. The linear bias of the n -variable function constructed in the last of this line of papers, i.e. in [13], is at most $(0.402963 + \ln 2/\pi)2^{-(n-2)/2} + \delta$, where δ is a small (though quite complicated) quantity. For odd n , Corollary 1 provides constructions of n -variable balanced functions with almost optimal linear bias of $2^{-(n-1)/2}$ and a lower bound of $\lceil (n-1)/4 \rceil$ on algebraic immunity. The linear bias of the functions obtained from the previous constructions (both the upper bound as well as actual values for concrete functions) are higher than the linear bias of the functions obtained from Corollary 1. So Corollary 1 and the previous constructions provide different points on the nonlinearity/linear bias trade-off curve for balanced functions.

The main advantage of the functions constructed using Corollary 1 is that these functions can be constructed using $O(n)$ gates, while all previous constructions are essentially based on discrete logarithm computation and require super-polynomial size circuits. To see this advantage in concrete terms, we consider the case of the 28-variable function (say f_{28}) reported in Table 2 of [13] which has algebraic immunity 14 and nonlinearity 134201460 (equivalently linear bias equal to about $2^{-13.01}$). According to the description of hardware implementation in Section 4.2 of [13], implementation of f_{28} will require a look-up table of size 2^{28} along with other gates. Taking $n = 57$ in Corollary 1, we obtain a function (say f_{57}) with algebraic immunity 14 and linear bias equal to 2^{-28} which can be implemented using 217 NAND, 28 XOR, 30 AND, and 1 OR gates (see [7] for the method of obtaining the gate count). Comparing f_{28} with f_{57} , we see that both are balanced and have the same algebraic immunity, f_{57} has a much lower linear bias, and can be implemented much more efficiently than f_{28} . From both security and efficiency points of view, to a designer of cryptosystems f_{57} will be much more preferable than f_{28} .

The special case of 1-resilient function obtained from Theorem 3 is given in the following result.

Corollary 2 *Let $n \equiv 0 \pmod{2}$. It is possible to construct an n -variable, 1-resilient function with linear bias equal to $2^{-(n-2)/2}$ (which is almost optimal), algebraic immunity at least $\lceil (n-2)/4 \rceil$, and can be implemented using $O(n)$ gates.*

Proof: Putting $m = 1$ in Theorem 3 provides the result. □

For even n , several papers [26, 29, 23, 30, 24] have proposed constructions of 1-resilient functions with optimal algebraic immunity $n/2$ and provable upper bounds on linear bias. To the best of our knowledge the last of such results is [24] which improves upon works prior to it by providing lower linear bias, and also guarantees the fast algebraic immunity to be at least $n - 6$. The upper bound on the linear bias of n -variable functions constructed in [24] is $((1 + (n/2) \ln 2)/\pi + (\pi + 16)/32)2^{-n/2+1} + 2^{-(n-1)}$, whereas the linear bias of the functions constructed in Corollary 2 is $2^{-(n-2)/2}$ which is almost optimal, the algebraic immunity is at least $\lceil (n-2)/4 \rceil$, and hence the fast algebraic immunity is at least $1 + \lceil (n-2)/4 \rceil$. So for 1-resilient functions, the functions constructed in [24] and those in Corollary 2 represent two distinct trade-off points with respect to algebraic immunity, fast algebraic immunity and linear bias. From an implementation point of view, however, there is a major difference between the constructions in Corollary 2 and that in [24]. The functions constructed using Corollary 2 require a circuit size of $O(n)$ gates, while the construction in [24] is based on defining the support of the desired Boolean function using powers of a primitive element over the field $\mathbb{F}_{2^{n/2}}$; in particular, from the description of Construction 2 in [24] it appears that the only reasonable method of implementing the function is to use a truth table requiring $\Omega(2^n)$ gates. We provide a concrete example to highlight the difference between the functions constructed using Corollary 2 and Theorem 6 of [24] (which is based on Construction 2 of [24]). Suppose $n = 32$. Theorem 6 of [24] provides a 32-variable, 1-resilient function (say g_{32})

with algebraic immunity 16, fast algebraic immunity at least 26 and nonlinearity at least 2147192232 (equivalently, linear bias at most about $2^{-12.85}$) which can be implemented using around 2^{32} gates. In Corollary 2 if we take $n = 102$, then we obtain a 102-variable, 1-resilient function (say g_{102}) with algebraic immunity at least 25, and hence fast algebraic immunity at least 26, and almost optimal linear bias equal to 2^{-50} which can be implemented using 406 NAND, 52 XOR, 52 AND, and 3 OR gates (see [7] for the method of obtaining the gate count). So g_{102} and g_{32} both are 1-resilient and have the same fast algebraic immunity, g_{102} has much higher algebraic immunity and much lower linear bias than g_{32} ; and g_{102} can be implemented much more efficiently than g_{32} . From both security and efficiency points of view, to a designer of cryptosystems g_{102} will be much more preferable than g_{28} .

Remark 1 *Theorem 3 guarantees algebraic immunity of the constructed n -variable, m -resilient function to be at least $\lceil (n - m - 1)/4 \rceil$. For concrete values of n , the actual value of algebraic immunity can actually be greater than the lower bound. For example, if we take $n = 10$ and $m = 1$, then the lower bound on algebraic immunity of the 10-variable function constructed using Theorem 3 is 2; we constructed the 10-variable function and computed its actual algebraic immunity which turns out to be 3.*

Theorem 3 covers the case where n and m do not have the same parity. The case where n and m have the same parity can also be covered in a similar manner. For such a construction we use a 5-variable function given by the following result.

Proposition 4 *Define*

$$f_5(X_1, X_2, Z_1, Z_2, Z_3) = Z_1 \oplus Z_2 \oplus X_1(Z_1 \oplus Z_3) \oplus X_2(Z_2 \oplus Z_3) \oplus X_1X_2(Z_1 \oplus Z_2 \oplus Z_3). \quad (6)$$

The function f_5 is a 5-variable, 1-resilient function having degree 3, algebraic immunity 2, nonlinearity 12 (and hence almost optimal linear bias equal to 2^{-2}), and can be implemented using 7 XOR and 4 AND gates.

Proof: We note that f_5 can be written in the following manner.

$$\begin{aligned} f_5(X_1, X_2, Z_1, Z_2, Z_3) = & (1 \oplus X_1)(1 \oplus X_2)(Z_1 \oplus Z_2) \oplus (1 \oplus X_1)X_2(Z_1 \oplus Z_3) \\ & \oplus X_1(1 \oplus X_2)(Z_2 \oplus Z_3) \oplus X_1X_2(Z_1 \oplus Z_2 \oplus Z_3). \end{aligned}$$

This shows that f_5 is the concatenation of 4 linear functions each of which is non-degenerate on at least 2 variables. It follows that f_5 is 1-resilient. It is easy to see that the degree of f_5 is 3. Further, it is not difficult to verify that the distance of f_5 to any affine is one of the values 12, 16 or 20, and so the nonlinearity of f_5 is 12. Hence, f_5 has almost optimal linear bias equal to 2^{-2} . Since the degree of f_5 is 3, its algebraic immunity is at most 3. It is easy to see that neither f_5 nor $1 \oplus f_5$ has any annihilator of degree 1. The following function is an annihilator of f_5 : $Z_1X_1 \oplus Z_1 \oplus Z_2 \oplus X_2 \oplus Z_2 \oplus Z_3 \oplus X_1 \oplus Z_3X_2 \oplus X_1X_2 \oplus 1$. So the algebraic immunity of f_5 is 2. The expression for f_5 given by (6) can be implemented using 7 XOR and 4 AND gates. $\square$

Using Proposition 4, we provide the construction for the case where n and m have the same parity.

Theorem 4 *Let m be a positive integer, and let $n \geq m + 4$ be such that $n \equiv m \pmod{2}$. Let $k = (n - m - 4)/2$. Let $\psi : \{0, 1\}^k \rightarrow \{0, 1\}^k$ be a bit permutation. Let f_5 be the function defined in (6). Define*

$$f(Y_1, \dots, Y_{m-1}, X_1, X_2, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k)$$

$$= Y_1 \oplus \cdots \oplus Y_{m-1} \oplus f_5(X_1, X_2, Z_1, Z_2, Z_3) \oplus (\psi, h)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k). \quad (7)$$

Then f is an n -variable, m -resilient function with linear bias equal to $2^{-(n-m)/2}$. Further, if $h = \text{Maj}_k$, then the algebraic immunity of f is at least $\lceil (n - m - 4)/4 \rceil$, and f can be implemented using $O(n)$ gates.

Proof: The function f_5 and $m - 1$ new variables are added to the bent function. The function f_5 is itself 1-resilient, and adding the $m - 1$ new variables increases resiliency to m . The linear bias of the bent function on $2k$ variables is 2^{-k} , where $k = (n - m - 4)/2$. The linear bias of f_5 is 2^{-2} , and so the linear bias of the sum of f_5 and the bent function is 2^{-k-2} . Addition of the new variables does not change the linear bias.

From Proposition 3 the algebraic immunity of f is at least the algebraic immunity of the bent function; from Theorem 2 the algebraic immunity of the bent function is at least the algebraic immunity of $h = \text{Maj}_k$; and from Theorem 1 the algebraic immunity of Maj_k is equal to $\lceil k/2 \rceil$. From Proposition 2, Maj_k can be implemented using $O(k) = O(n)$ gates. The implementation of the other components of f also require $O(n)$ gates. $\square$

4 Iterated Construction

Both Theorems 3 and 4 essentially simply add new variables to increase resiliency. This may be considered undesirable. In this section, we describe a different method for increasing resiliency. To do this, we resurrect an idea of an iterated construction of resilient functions which was only briefly sketched in [17]. The idea in [17] was itself based on a more general theoretical result from [25]. The description in [17] briefly considered resiliency and nonlinearity, but not algebraic immunity (in fact, the work [17] predates the introduction of the notion of algebraic immunity).

Construction 2 Let g and h be two n -variable functions, and let f be an $(n + 1)$ -variable function obtained as $\text{Concat}(g, h)$, i.e.

$$f(X_1, \dots, X_{n+1}) = (1 \oplus X_{n+1})g(X_1, \dots, X_n) \oplus X_{n+1}h(X_1, \dots, X_n).$$

Define $(n + 3)$ -variable functions G and H as follows.

$$\begin{aligned} G(X_1, \dots, X_{n+3}) &= X_{n+3} \oplus X_{n+2} \oplus f(X_1, \dots, X_{n+1}) \\ &= X_{n+3} \oplus X_{n+2} \oplus (1 \oplus X_{n+1})g(X_1, \dots, X_n) \oplus X_{n+1}h(X_1, \dots, X_n), \end{aligned} \quad (8)$$

$$\begin{aligned} H(X_1, \dots, X_{n+3}) &= X_{n+3} \oplus X_{n+1} \oplus (1 \oplus X_{n+3} \oplus X_{n+2})g(X_1, \dots, X_n) \oplus (X_{n+3} \oplus X_{n+2})h(X_1, \dots, X_n). \end{aligned} \quad (9)$$

By $\text{Step}(g, h)$ we denote the pair of functions (G, H) obtained in (8) and (9). Define an $(n + 4)$ -variable function F as $\text{Concat}(G, H)$, i.e.

$$F(X_1, \dots, X_{n+4}) = (1 \oplus X_{n+4})G(X_1, \dots, X_{n+3}) \oplus X_{n+4}H(X_1, \dots, X_{n+3}). \quad (10)$$

The following result relates the properties of g and h to that of G , H and F .

Theorem 5 Let n be a positive integer, and g and h be two n -variable functions. Let $(G, H) = \text{Step}(g, h)$ and $F = \text{Concat}(G, H)$ be constructed as in Construction 2. Then the following holds.

1. If g and h are m -resilient, then G and H are $(m+2)$ -resilient.
2. $\text{nl}(G) = \text{nl}(H) = 4\text{nl}(f)$, and equivalently $\text{LB}(G) = \text{LB}(H) = \text{LB}(f)$.
3. Let $\ell(X_1, \dots, X_{n+3})$ be a linear function. Then either $G \oplus \ell$ or $H \oplus \ell$ is balanced.
4. Obtaining G and H from g and h requires 8 XOR and 4 AND gates.
5. F is $(m+2)$ -resilient.
6. $\text{nl}(F) = 2^{n+2} + \text{nl}(G) = 2^{n+2} + 4\text{nl}(f)$, and equivalently $\text{LB}(F) = \text{LB}(G)/2 = \text{LB}(f)/2$.
7. Obtaining F from G and H requires 2 XOR and 2 AND gates.

Proof: If g and h are m -resilient, then so is f . Note that G is obtained by adding two new variables to f . It then follows that G is $(m+2)$ -resilient and further $\text{nl}(G) = 4\text{nl}(f)$. The function H is obtained from the function G by the following invertible linear transformation on the variables: $X_{n+1} \rightarrow X_{n+3} \oplus X_{n+2}$, $X_{n+2} \rightarrow X_{n+1}$, $X_{n+3} \rightarrow X_{n+3}$. Since nonlinearity is invariant under an invertible linear transformation on the variables, it follows that $\text{nl}(H) = \text{nl}(G)$.

The function $H(X_1, \dots, X_{n+3})$ can be written as $H(X_1, \dots, X_{n+3}) = X_{n+1} \oplus H'(X_1, \dots, X_n, X_{n+2}, X_{n+3})$, where

$$\begin{aligned} H'(X_1, \dots, X_n, X_{n+2}, X_{n+3}) \\ = (1 \oplus X_{n+3})(1 \oplus X_{n+2})g(X_1, \dots, X_n) \oplus (1 \oplus X_{n+3})X_{n+2}h(X_1, \dots, X_n) \\ \oplus X_{n+3}(1 \oplus X_{n+2})(1 \oplus h(X_1, \dots, X_n)) \oplus X_{n+3}X_{n+2}(1 \oplus g(X_1, \dots, X_n)). \end{aligned}$$

Since H is obtained from H' by adding a new variable, it follows that to show H is $(m+2)$ -resilient, it is sufficient to show H' is $(m+1)$ -resilient. To show that H' is $(m+1)$ -resilient, it is sufficient to show that $H' \oplus \ell'$ is balanced for all $(n+2)$ -variable linear functions $\ell'(X_1, \dots, X_n, X_{n+2}, X_{n+3})$ which are non-degenerate on at most $m+1$ variables. Let ℓ' be any such linear function. Write

$$\begin{aligned} \ell'(X_1, \dots, X_n, X_{n+2}, X_{n+3}) \\ = (1 \oplus X_{n+3})(1 \oplus X_{n+2})\ell_1(X_1, \dots, X_n) \oplus (1 \oplus X_{n+3})X_{n+2}\ell_2(X_1, \dots, X_n) \\ \oplus X_{n+3}(1 \oplus X_{n+2})\ell_3(X_1, \dots, X_n) \oplus X_{n+3}X_{n+2}\ell_4(X_1, \dots, X_n), \end{aligned}$$

where $\ell_1, \ell_2, \ell_3, \ell_4$ are linear functions. Note that for $1 \leq i < j \leq 4$, either $\ell_i = \ell_j$ or $\ell_i = 1 \oplus \ell_j$. We have

$$\text{wt}(H' \oplus \ell') = \text{wt}(g \oplus \ell_1) + \text{wt}(h \oplus \ell_2) + (2^n - \text{wt}(h \oplus \ell_3)) + (2^n - \text{wt}(g \oplus \ell_4)). \quad (11)$$

Suppose that ℓ' is degenerate on both X_{n+2} and X_{n+3} . In this case, all the ℓ_i 's are equal, and so from (11) it follows that $\text{wt}(H' \oplus \ell) = 2^{n+1}$. Next suppose that ℓ is non-degenerate on at least one of X_{n+2} or X_{n+3} . In this case, each of the ℓ_i 's is non-degenerate on at most m variables. Since both g and h are m -resilient, it follows that $\text{wt}(g \oplus \ell_1)$, $\text{wt}(h \oplus \ell_2)$, $\text{wt}(h \oplus \ell_3)$ and $\text{wt}(g \oplus \ell_4)$ are all equal to 2^{n-1} , and so $\text{wt}(H' \oplus \ell) = 2^{n+1}$. This shows that H' is $(m+1)$ -resilient and hence H is $(m+2)$ -resilient.

Now we consider the proof of the third point. Suppose the linear function $\ell(X_1, \dots, X_{n+3})$ is degenerate X_{n+3} . Then $G \oplus \ell$ can be written as X_{n+3} plus a function which does not involve X_{n+3} , and hence $G \oplus \ell$ is balanced. Similarly, if ℓ is degenerate on X_{n+2} , then $G \oplus \ell$ can be written as X_{n+2} plus a function which does not involve X_{n+2} , and hence $G \oplus \ell$ is balanced. Further, if ℓ is degenerate on X_{n+1} , then $H \oplus \ell$ can be written as X_{n+1} plus a function which does not involve X_{n+1} , and hence $H \oplus \ell$ is balanced. So suppose ℓ is non-degenerate on all three of the variables X_{n+1}, X_{n+2} and X_{n+3} , i.e. $\ell(X_1, \dots, X_{n+3}) = X_{n+3} \oplus X_{n+2} \oplus X_{n+1} \oplus \sigma(X_1, \dots, X_n)$, where σ is a linear function. We argue that $H \oplus \ell$ is balanced. From the definition of H , we have $H \oplus \ell = X_{n+2} \oplus (1 \oplus X_{n+3} \oplus X_{n+2})g \oplus (X_{n+3} \oplus X_{n+2})h \oplus \sigma$

(which is degenerate on X_{n+1}). Considering the four possible values of X_{n+3} and X_{n+2} , the sub-functions of $H \oplus \ell$ that are obtained are $g \oplus \sigma$ (corresponding to $X_{n+3} = 0, X_{n+2} = 0$), $1 \oplus h \oplus \sigma$ (corresponding to $X_{n+3} = 0, X_{n+2} = 1$), $h \oplus \sigma$ (corresponding to $X_{n+3} = 1, X_{n+2} = 0$), and $1 \oplus g \oplus \sigma$ (corresponding to $X_{n+3} = 1, X_{n+2} = 1$). So $\text{wt}(H \oplus \ell) = 2(\text{wt}(g \oplus \sigma) + \text{wt}(1 \oplus h \oplus \sigma) + \text{wt}(h \oplus \sigma) + \text{wt}(1 \oplus g \oplus \sigma)) = 2^{n+2}$, where the factor of 2 arises from the variable X_{n+1} on which $H \oplus \ell$ is degenerate. Hence, $H \oplus \ell$ is balanced.

The counts of the XOR and AND gates are clear from (8) and (9).

Next we provide the arguments for the properties of F . Since H and G are $(m+2)$ -resilient, it follows that F is also $(m+2)$ -resilient. Let $\mu(X_1, \dots, X_{n+4})$ be an affine function on $(n+4)$ variables. Then μ can be written as $\mu = c \oplus dX_{n+4} \oplus \ell(X_1, \dots, X_{n+3})$, where $c, d \in \mathbb{F}_2$, and ℓ is a linear function on $n+3$ variables. Then $\text{wt}(F \oplus \mu) = \text{wt}(c \oplus G \oplus \ell) + \text{wt}(c \oplus d \oplus H \oplus \ell)$. From the third point of the theorem, we have that either $G \oplus \ell$, or $H \oplus \ell$ is balanced. Suppose $G \oplus \ell$ is balanced. Then $\text{wt}(F \oplus \mu) = 2^{n+2} + \text{wt}(c \oplus d \oplus H \oplus \ell) \geq 2^{n+2} + \text{nl}(H)$, where the equality is achieved for c, d and ℓ such that $\text{nl}(H) = \text{wt}(c \oplus d \oplus H \oplus \ell)$. Similarly, if $H \oplus \ell$ is balanced, then $\text{wt}(F \oplus \mu) = 2^{n+2} + \text{wt}(c \oplus G \oplus \ell)$, where the equality is achieved for c and ℓ such that $\text{nl}(G) = \text{wt}(c \oplus G \oplus \ell)$. From these two cases, the statement on the nonlinearity of F follows. $\square$

To obtain a lower bound on the algebraic immunities of G and H obtained from Construction 1, we first prove the following general result.

Proposition 5 *Let n, n_1 and n_2 be positive integers with $n = n_1 + n_2$, and let $f(X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2})$ be an n -variable function. We write*

$$f(X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2}) = \bigoplus_{\alpha=(\alpha_1, \dots, \alpha_{n_2})} (1 \oplus \alpha_1 \oplus Y_1) \cdots (1 \oplus \alpha_{n_2} \oplus Y_{n_2}) f_\alpha(X_1, \dots, X_{n_1}), \quad (12)$$

where for $\alpha \in \mathbb{F}_2^{n_2}$, $f_\alpha(X_1, \dots, X_{n_1}) = f(X_1, \dots, X_{n_1}, \alpha_1, \dots, \alpha_{n_2})$. Then $\text{Al}(f) \geq \min_{\alpha \in \mathbb{F}_2^{n_2}} \text{Al}(f_\alpha)$.

Proof: Suppose g is a non-zero annihilator of f . We write

$$g(X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2}) = \bigoplus_{\alpha=(\alpha_1, \dots, \alpha_{n_2})} (1 \oplus \alpha_1 \oplus Y_1) \cdots (1 \oplus \alpha_{n_2} \oplus Y_{n_2}) g_\alpha(X_1, \dots, X_{n_1}),$$

where $g_\alpha(X_1, \dots, X_{n_1}) = g(X_1, \dots, X_{n_1}, \alpha_1, \dots, \alpha_{n_2})$.

Since $gf = 0$, it follows that $g_\alpha f_\alpha = 0$ for all $\alpha \in \mathbb{F}_2^{n_2}$. Further, since $g \neq 0$, there must be some α such that $g_\alpha \neq 0$. Then g_α is a non-zero annihilator of f_α and so $\deg(g_\alpha) \geq \text{Al}(f_\alpha)$. Clearly, $\deg(g) \geq \deg(g_\alpha)$, and so $\deg(g) \geq \text{Al}(f_\alpha)$.

A similar reasoning shows that if g is a non-zero annihilator of $1 \oplus f$, then $\deg(g) \geq \text{Al}(f_\beta)$ for some $\beta \in \mathbb{F}_2^{n_2}$.

Since $\text{Al}(f)$ is the minimum of the degrees of all the non-zero annihilators of f , and the degrees of all the non-zero annihilators of $1 \oplus f$, the result follows. $\square$

The lower bound on the algebraic immunity of a direct sum which was obtained in [15] and is stated in Proposition 3 can be seen as a corollary of Proposition 5. The idea is that the sub-functions of f obtained by fixing $X_1, \dots, X_{n_1}$ to arbitrary values are either $h(Y_1, \dots, Y_{n_2})$ or $1 \oplus h(Y_1, \dots, Y_{n_2})$. So from Proposition 5, we have $\text{Al}(f) \geq \text{Al}(h)$. Similarly, by fixing $Y_1, \dots, Y_{n_2}$ to arbitrary values, we obtain $\text{Al}(f) \geq \text{Al}(g)$.

Theorem 6 Let G , H and F be the functions constructed as in Construction 2. Then

1. $\text{Al}(G), \text{Al}(H) \geq \min\{\text{Al}(g), \text{Al}(h)\}$.
2. $\text{Al}(F) \geq \min\{\text{Al}(G), \text{Al}(H)\} \geq \min\{\text{Al}(g), \text{Al}(h)\}$.

Proof: By setting the variables $X_{n+3}, X_{n+2}, X_{n+1}$ to arbitrary values, the sub-functions of G that are obtained are g , $1 \oplus g$, h , and $1 \oplus h$. Since g and $1 \oplus g$ have the same algebraic immunity, and h and $1 \oplus h$ also have the same algebraic immunity, the lower bound on $\text{Al}(G)$ follows from Proposition 5. The lower bounds on $\text{Al}(H)$ and $\text{Al}(F)$ follow in a similar manner using Proposition 5. $\square$

Construction 2 can be iterated to obtain functions on progressively larger number of variables. The following construction describes the idea.

Construction 3 Let n and t be positive integers, and let g and h be two n -variable functions. Consider the following iterated construction.

```

 $g^{(0)} \leftarrow g; h^{(0)} \leftarrow h; f^{(0)} \leftarrow \text{Concat}(g^{(0)}, h^{(0)});$ 
for  $i \leftarrow 1$  to  $t$  do
     $(g^{(i)}, h^{(i)}) \leftarrow \text{Step}(g^{(i-1)}, h^{(i-1)}); f^{(i)} \leftarrow \text{Concat}(g^{(i)}, h^{(i)});$ 
end for;
return  $f^{(t)}$ .

```

We denote the function $f^{(t)}$ by $\text{lter}_t(g, h)$.

The properties of the function constructed using Construction 3 are given in the following result.

Theorem 7 Let n and t be positive integers, g and h be n -variable functions. Let $f = \text{Concat}(g, h)$, and $f^{(t)} = \text{lter}_t(g, h)$. Then the following holds.

1. The function $f^{(t)}$ is an $(n + 3t + 1)$ -variable function.
2. If g and h are m -resilient, then $f^{(t)}$ is $(m + 2t)$ -resilient.
3. $\text{LB}(f^{(t)}) = 2^{-t} \cdot \text{LB}(f)$.
4. $\text{Al}(f^{(t)}) \geq \min\{\text{Al}(g), \text{Al}(h)\}$.
5. Obtaining $f^{(t)}$ from g and h requires $8t + 2$ XOR and $4t + 2$ AND gates.

Proof: The result follows from Theorem 5 by induction on t . $\square$

By appropriately choosing the initial functions g and h , Theorem 7 can be used to obtain the resiliency, linear bias and lower bound on algebraic immunity of the function $\text{lter}_t(g, h)$. This is stated in the following result.

Theorem 8 Let n and t be positive integers.

1. Let $n - 3t - 1 \geq 3$ be an odd integer. Let $k = (n - 3t - 2)/2$, and $\psi : \{0, 1\}^k \rightarrow \{0, 1\}^k$ be a bit permutation. Define

$$\begin{aligned}
 g^{(0)}(X_1, U_1, \dots, U_k, V_1, \dots, V_k) &= X_1 \oplus (\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k) \\
 h^{(0)}(X_1, U_1, \dots, U_k, V_1, \dots, V_k) &= 1 \oplus X_1 \oplus (\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k) \\
 f^{(t)} &= \text{lter}_t(g^{(0)}, h^{(0)}).
 \end{aligned}$$

Then $f^{(t)}$ is an n -variable, $2t$ -resilient function with linear bias equal to $2^{-(n-t-2)/2}$, algebraic immunity at least $\lceil (n - 3t - 2)/4 \rceil$, and can be implemented using $O(n)$ gates.

2. Let $n - 3t - 1 \geq 4$ be an even integer. Let $k = (n - 3t - 5)/2$, and $\psi : \{0, 1\}^k \rightarrow \{0, 1\}^k$ be a bit permutation. Define

$$\begin{aligned} g^{(0)}(X_1, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k) \\ &= Z_1 \oplus Z_2 \oplus X_1(Z_1 \oplus Z_3) \oplus (\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k) \\ h^{(0)}(X_1, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k) \\ &= Z_1 \oplus Z_3 \oplus X_1 Z_2 \oplus (\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k) \\ f^{(t)} &= \text{Iter}_t(g^{(0)}, h^{(0)}). \end{aligned}$$

Then $f^{(t)}$ is an n -variable, $(2t+1)$ -resilient function with linear bias equal to $2^{-(n-t-1)/2}$, algebraic immunity at least $\lceil (n - 3t - 5)/4 \rceil$, and can be implemented using $O(n)$ gates.

Proof: First suppose $n - 3t - 1$ is odd.

Note that due to the addition of the variable X_1 , both $g^{(0)}$ and $h^{(0)}$ are balanced, i.e. 0-resilient. From Proposition 3, the algebraic immunities of both $g^{(0)}$ and $h^{(0)}$ are at least the algebraic immunity of $(\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k)$; from Theorem 2 the algebraic immunity of $(\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k)$ is at least the algebraic immunity of Maj_k ; and from Theorem 1, the algebraic immunity of Maj_k is at least $\lceil k/2 \rceil = \lceil (n - 3t - 2)/4 \rceil$. So the algebraic immunities of both $g^{(0)}$ and $h^{(0)}$ are at least $\lceil (n - 3t - 2)/4 \rceil$. Define

$$\begin{aligned} f^{(0)}(X_1, X_2, U_1, \dots, U_k, V_1, \dots, V_k) \\ &= (1 \oplus X_2)g^{(0)}(X_1, U_1, \dots, U_k, V_1, \dots, V_k) \oplus X_2h^{(0)}(X_1, U_1, \dots, U_k, V_1, \dots, V_k). \end{aligned}$$

Simplifying we have $f^{(0)}(X_1, X_2, U_1, \dots, U_k, V_1, \dots, V_k) = X_1 \oplus X_2 \oplus (\psi, \text{Maj}_k)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k)$. From Theorem 3, the linear bias of $f^{(0)}$ is equal to $2^{-(n-3t-2)/2}$. Note that $g^{(0)}$ and $h^{(0)}$ are functions of $n - 3t - 1$ variables and $f^{(0)}$ is a function of $n - 3t$ variables. Since $f^{(t)} = \text{Iter}_t(g^{(0)}, h^{(0)})$, from Theorem 7 it follows that $f^{(t)}$ is a function of n variables. Further, also from Theorem 7, the resiliency of $f^{(t)}$ is $2t$, linear bias is equal to $2^{-t}\text{LB}(f^{(0)}) = 2^{-(n-t-2)/2}$, and algebraic immunity is at least $\lceil (n - 3t - 2)/4 \rceil$. The implementation of $f^{(t)}$ requires the implementation of Maj_k , and the implementation of the bit permutation ψ . The implementation of the bit permutation ψ does not require any gates (only the appropriate connection pattern needs to be implemented). From Proposition 2, Maj_k can be implemented using $O(k) = O(n)$ gates. From Theorem 7 obtaining $f^{(t)}$ from $g^{(0)}$ and $h^{(0)}$ requires $O(t) = O(n)$ gates. So overall $f^{(t)}$ can be implemented using $O(n)$ gates.

Next suppose $n - 3t - 1$ is even. By an argument similar to the case for $n - 3t - 1$ is odd, the algebraic immunities of both $g^{(0)}$ and $h^{(0)}$ are at least $\lceil k/2 \rceil = \lceil (n - 3t - 5)/4 \rceil$. Define

$$\begin{aligned} f^{(0)}(X_1, X_2, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k) \\ &= (1 \oplus X_2)g^{(0)}(X_1, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k) \\ &\quad \oplus X_2h^{(0)}(X_1, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k). \end{aligned}$$

Simplifying we have

$$\begin{aligned} f^{(0)}(X_1, X_2, Z_1, Z_2, Z_3, U_1, \dots, U_k, V_1, \dots, V_k) \\ &= f_5(X_1, X_2, Z_1, Z_2, Z_3) \oplus (\psi, h)\text{-MM}_{2k}(U_1, \dots, U_k, V_1, \dots, V_k). \end{aligned}$$

From Theorem 4, $f^{(0)}$ is 1-resilient. Note that $Z_1 \oplus Z_2 \oplus X_1(Z_1 \oplus Z_3) = (1 \oplus X_1)(Z_1 \oplus Z_2) \oplus X_1(Z_2 \oplus Z_3)$, which is the concatenation of two linear functions both of which are non-degenerate on 2 variables, and

hence is 1-resilient. Since $g^{(0)}$ is the direct sum of the 1-resilient function $Z_1 \oplus Z_2 \oplus X_1(Z_1 \oplus Z_3)$ and a bent function, it follows that $g^{(0)}$ is 1-resilient. Similarly, we may write $Z_1 \oplus Z_3 \oplus X_1 Z_2 = (1 \oplus X_1)(Z_1 \oplus Z_3) \oplus X_1(Z_1 \oplus Z_2 \oplus Z_3)$ to see that $h^{(0)}$ is the direct sum of a 1-resilient function and a bent function, and so it follows that $h^{(0)}$ is 1-resilient. From Theorem 4, the linear bias of $f^{(0)}$ is equal to $2^{-(2k+6-2)/2} = 2^{-(n-3t-1)/2}$. As in the case for $n - 3t - 1$ being odd, from Theorem 7, $f^{(t)}$ is a function of n variables, which is $(2t+1)$ -resilient, having linear bias equal to $2^{-t} \text{LB}(f^{(0)}) = 2^{-(n-t-1)/2}$, algebraic immunity at least $\lceil (n - 3t - 5)/4 \rceil$, and can be implemented using $O(n)$ gates. $\square$

5 Achieving Provable Resiliency/Nonlinearity/Algebraic Immunity Trade-Offs

In this section, we show new provable trade-offs between resiliency, nonlinearity and algebraic immunity. The precise result that we present is quite powerful. Suppose m_0 , x_0 , and a_0 are given. We show that it is possible to construct an n -variable function which is at least m_0 -resilient, has linear bias at most 2^{-x_0} , algebraic immunity at least a_0 , and can be implemented using $O(n)$ gates, where the number of variables n depends linearly on m_0 , x_0 and a_0 . As far as we are aware, there is no such comparable result in the literature.

In concrete terms, based on Theorems 3, 4 and 8, the following result states how to achieve desired target values of the order of resiliency, linear bias, and algebraic immunity.

Theorem 9 *Let m_0 be a non-negative integer, x_0 and a_0 be positive integers. The following holds.*

1. *Let $n \geq m_0 + 1 + 2 \cdot \max\{2a_0, x_0 - 1\}$. Then it is possible to construct an n -variable function whose resiliency order is m_0 , linear bias is at most 2^{-x_0} , algebraic immunity is at least a_0 .*
2. *Let $n \geq m_0 + 5 + 2 \cdot \max\{2a_0, x_0 - 3\}$. Then it is possible to construct an n -variable function whose resiliency order is m_0 , linear bias is at most 2^{-x_0} , algebraic immunity is at least a_0 .*
3. *Let $t = \lceil m_0/2 \rceil$ and $n \geq \max\{2x_0 + t, 4a_0 + 3t + 2\}$ be such that $n - 3t - 1$ is odd. Then it is possible to construct an n -variable function whose resiliency order is $2t \geq m_0$, linear bias is at most 2^{-x_0} , algebraic immunity is at least a_0 .*
4. *Let $t = \lceil (m_0 - 1)/2 \rceil$ and $n \geq \max\{2x_0 + t - 1, 4a_0 + 3t + 5\}$ be such that $n - 3t - 1$ is even. Then it is possible to construct an n -variable function whose resiliency order is $2t + 1 \geq m_0$, linear bias is at most 2^{-x_0} , algebraic immunity is at least a_0 .*

Further, in all of the above cases, the respective functions can be implemented using $O(n)$ gates.

Proof: The first two points of the theorem follow from Theorems 3 and 4 respectively. The last two points follow from the two corresponding points of Theorem 8. The statement regarding the number of gates also follows from Theorems 3, 4, and 8. $\square$

Theorem 10 *Let m_0 be a non-negative integer, x_0 and a_0 be positive integers. Any function which is at least m_0 -resilient, has linear bias at most 2^{-x_0} and algebraic immunity at least a_0 requires at least $\max(m_0 + 1, 2x_0, 2a_0 - 1) - 1$ gates for implementation.*

Consequently, in all the four cases of Theorem 9, the n -variable functions with the least value of n requires an asymptotically optimal number of gates for achieving the target values of m_0 , x_0 and a_0 .

target (m_0, x_0, a_0)	achieved			
	Thm 3 (n, m, x, a)	Thm 4 (n, m, x, a)	Thm 8(1) (n, m, x, a)	Thm 8(2) (n, m, x, a)
(4,6,3)	(17,4,6,3)	(20,4,8,3)	(20,4,8,3)	(23,5,10,3)
(4,6,4)	(21,4,8,4)	(24,4,10,4)	(24,4,10,4)	(27,5,12,4)
(4,9,3)	(23,4,9,5)	(22,4,9,4)	(22,4,9,4)	(23,5,10,3)
(4,9,4)	(23,4,9,5)	(24,4,10,4)	(24,4,10,4)	(27,5,12,4)
(4,12,3)	(29,4,12,6)	(28,4,12,5)	(28,4,12,5)	(27,5,12,4)
(4,12,4)	(29,4,12,6)	(28,4,12,5)	(28,4,12,5)	(27,5,12,4)
(7,6,3)	(20,7,6,3)	(23,7,8,3)	(26,8,10,3)	(26,7,11,3)
(7,6,4)	(24,7,8,4)	(27,7,10,4)	(30,8,12,4)	(30,7,13,4)
(7,9,3)	(26,7,9,3)	(25,7,9,4)	(26,8,10,3)	(26,7,11,3)
(7,9,4)	(26,7,9,5)	(27,7,10,4)	(30,8,12,4)	(30,7,13,4)
(7,12,3)	(32,7,12,6)	(31,7,12,5)	(30,8,12,4)	(28,7,12,4)
(7,12,4)	(32,7,12,6)	(31,7,12,5)	(30,8,12,4)	(30,7,13,4)

Table 1: Examples of trade-offs achieved by the various constructions.

Proof: Any m_0 -resilient function is non-degenerate on at least $m_0 + 1$ variables, any function with linear bias at most 2^{-x_0} is non-degenerate on at least $2x_0$ variables, and any function with algebraic immunity at least a_0 is non-degenerate on at least $2a_0 - 1$ variables. The first statement now follows from Proposition 1. To see the second statement, note that in all the four cases of Theorem 9, the lower bound on n is linear in $\max(m_0, x_0, a_0)$. $\square$

Given m_0 , x_0 , and a_0 , each of the four points of Theorem 9 provides infinitely many values of n achieving the desired properties. From an implementation point of view, for each of the four points, one would choose the smallest value of n satisfying the stated conditions. This is given by the lower bounds on n for the different cases. The trade-offs achieved by the constructions in Theorems 3, 4 and 8 and summarised in Theorem 9 are different. No one construction can be said to subsume one of the other. In Table 1, we provide some examples to illustrate this point. For each target value (m_0, x_0, a_0) , the table provides (n, m, x, a) for the various constructions, where n is the smallest number of variables which guarantees the target values, while (m, x, a) are the actual values that are achieved.

6 Conclusion

We have described several constructions which provide functions with provable trade-offs between resiliency, linear bias, and algebraic immunity. As far as we are aware there is no previous work in the literature which addresses the trade-off question in the same generality that we do. The constructions that we describe are simple and provide functions which can be efficiently implemented. Our work opens the possibility of several promising directions of new research. One direction is to obtain constructions which achieve better provable trade-offs between resiliency, linear bias and algebraic immunity. From a practical cryptographic point of view, it would be good to keep implementation efficiency in mind while obtaining new trade-offs. While the functions that we have described require an asymptotically optimal number of gates, in concrete terms the possibility of obtaining functions which can be implemented with even smaller number of gates remain open. We hope that these questions will be of interest to the Boolean function research community and lead to new results in the future.

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